

Daniel Rakita



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🎓 Google Scholar

Curriculum Vitae — June 2026

Research Mission

My research primarily involves formulating planning, optimization, and learning algorithms that allow robot manipulation platforms to effectively complete tasks. The goal of my work is to enable people to intuitively control or work alongside robot manipulation platforms to perform critical tasks deemed unsuitable, undesirable, understaffed, or unsafe for people, such as full-time homecare, home assistance, telenursing, robot surgery, disaster relief, large-scale manufacturing, nuclear materials handling, and space robotics. I use interdisciplinary techniques across robotics and computer science, including motion planning, motion optimization, shared autonomy, human-robot interaction, and machine learning to formulate and validate generalizable, end-to-end solutions within these problem spaces.

Appointments

2022 - Current	Assistant Professor Yale University, Department of Computer Science
2015-2022	Graduate Researcher University of Wisconsin-Madison Visual Computing Lab and Human-Computer Interaction Lab
2018, 2019	NREIP Researcher Naval Research Lab, Washington, D.C.

Education

2015-2022	Ph.D. in Computer Science University of Wisconsin-Madison <i>Advisors: Michael Gleicher and Bilge Mutlu</i>
2015-2017	Masters of Computer Science University of Wisconsin-Madison
2013-2015	Undergraduate work in computer science University of Wisconsin-Madison
2008-2012	Bachelor of Music in Performance Indiana University-Bloomington Jacobs School of Music

Selected Honors Awards

2023	Best Paper Award Winner, <i>ACM/IEEE Conference on Human-Robot Interaction (HRI)</i>
2022	Outstanding Graduate-Student Research Award, <i>UW-Madison</i>
2021	Outstanding Reviewer Award, <i>Selected by IROS Conference Paper Review Board, Top 4 of 3,942</i>
2021	Cisco Graduate Student Fellowship Recipient, <i>UW-Madison</i>
2021	Three Minute Thesis Competition Finalist, <i>UW-Madison</i>
2020	Best Paper Award Finalist, <i>ACM/IEEE Conference on Human-Robot Interaction (HRI)</i>
2019	Microsoft PhD Fellowship Recipient
2018	Best Paper Award Winner, <i>ACM/IEEE Conference on Human-Robot Interaction (HRI), Top 4 of 206 papers</i>
2017	NSF Graduate Research Fellowship Program Honorable Mention

Selected Research Grants and Funding

2025 - 2026	Advanced AI-Based Manipulation for USV Diesel Engine Maintenance (Office of Naval Research) Role: Key Investigator, Amount: \$160,000 <i>STTR Phase 1. Key Investigators: Brian Kononchik (Boston Engineering), Daniel Rakita (Yale), Brian Scassellati (Yale), John Steagall (Fairbanks Morse Defense).</i>
2024 - 2027	Robot Manipulation in Densely Cluttered Environments (Office of Naval Research) Role: Principal Investigator, Amount: \$1,212,027 <i>Co-PI(s): Brian Scassellati.</i>
2021-2022	Cisco Graduate Student Fellowship (Cisco) Amount: one year PhD tuition and stipend
2019-2020	Microsoft PhD Fellowship (Microsoft) Amount: \$84,000 for tuition, stipend, and travel funds

Selected Publications

Conference

2026	Sun, X., Wang, Y., Yang, S., Chen, Y., and Rakita, D. Hybrid Diffusion Policies with Projective Geometric Algebra for Efficient Robot Manipulation Learning. <i>ICRA</i> . [Link]  Oral Presentation Liang, C., Sun, X., Wang, Q., and Rakita, D. Turning Stale Gradients into Stable Gradients: Coherent Coordinate Descent with Implicit Landscape Smoothing for Lightweight Zeroth-Order Optimization. <i>ICML</i> . Wang, Q., Abdellall, O., Gao, T., Sun, X., and Rakita, D. Subsecond 3D Mesh Generation for Robot Manipulation. <i>ICRA</i> . [Link]
2025	Liang, C., Wang, P., Xu, A., and Rakita, D. ad-trait: A Fast and Flexible Automatic Differentiation Library in Rust. <i>IROS</i> . [Link] Hoffmeister, L., Scassellati, B., and Rakita, D. Towards Zero-Knowledge Task Planning via a Language-based Approach. <i>IROS</i> . Rakita, D. , Liang, C., and Wang, Q. Coherence-based Approximate Derivatives via Web of Affine Spaces Optimization. <i>RSS</i> . [Link] Sun, X., Yang, S., Chen, Y., Fan, F., Liang, Y., and Rakita, D. Dynamic Rank Adjustment in Diffusion Policies for Efficient and Flexible Training. <i>RSS</i> . [Link]
2024	Hoffmeister, L., Scassellati, B., and Rakita, D. Sequential Discrete Action Selection via Blocking Conditions and Resolutions. <i>IROS</i> . [Link]
2023	Schoen, A., Sullivan, D., Zhang, Z., Rakita, D. , and Mutlu, B. Lively: Enabling Multimodal, Lifelike, and Extensible Real-time Robot Motion. <i>HRI</i> . [Link]  Best Paper Award Winner Patel, V., Rakita, D. , and Dollar, A. An Analysis of Unified Manipulation with Robot Arms and Dexterous Hands via Optimization-based Motion Synthesis. <i>ICRA</i> . [Link] Wang, Y., Praveena, P., Rakita, D. , and Gleicher, M. RangedIK: An Optimization-based Robot Motion Generation Method for Ranged-Goal Tasks. <i>ICRA</i> . [Link]
2022	Rakita, D. , Mutlu, B., and Gleicher, M. PROXIMA: An Approach for Time or Accuracy Budgeted Collision Proximity Queries. <i>RSS</i> . [Link]
2021	Rakita, D. , Shi, H., Mutlu, B., and Gleicher, M. CollisionIK: A Per-Instant Pose Optimization Method for Generating Robot Motions. <i>ICRA</i> . Rakita, D. , Mutlu, B., and Gleicher, M. Strobe: An Acceleration Meta-algorithm for Optimizing Robot Paths using Concurrent Interleaved Sub-Epoch Pods. <i>ICRA</i> .
2020	Praveena, P., Rakita, D. , Mutlu, B., and Gleicher, M. Supporting Perception of Weight through Motion-induced Sensory Conflicts in Robot Teleoperation. <i>HRI</i> .  Best Paper Award Nominee
2018	Rakita, D. , Mutlu, B., and Gleicher, M. An Autonomous Dynamic Camera Method for Effective Remote Teleoperation. <i>HRI</i> .  Best Paper Award Winner

- 2016 Bodden, C., **Rakita, D.**, Mutlu, B., and Gleicher, M. Evaluating Intent-Expressive Robot Arm Motion. *RO-MAN*.
 **Best Paper Nominee**

Journal

- 2026 Bader, J., Sun, X., Rosenfelt, T., Ramirez-Hardy, A., Vitchutripop, T., Pantel, H., Khanna, A., and **Rakita, D.** Deep learning approach for critical exposure during division of the inferior mesenteric artery in colorectal surgery. *Journal of Robotic Surgery*.[\[Link\]](#)
- 2021 Chamzas, C., Quintero-Pena, C., Kingston, Z., Orthey, A., **Rakita, D.**, Gleicher, M., Toussaint, M., and Kavraki, L. MotionBench-Maker: A Tool to Generate and Benchmark Motion Planning Datasets. *IEEE Robotics and Automation Letters*.[\[Link\]](#)
- Rakita, D.**, Mutlu, B., and Gleicher, M. Single Query Path Planning using Sample Efficient Probability Informed Trees. *RA-L/ICRA*.

Selected Invited Talks

- 2025 *The Applied Planning, Learning, and Optimization Toolbox: A Flexible Rust-Based Software Suite for Robotics Research*. Rust for Robotics Workshop, ICRA
- 2022 *Intuitive Robot Shared-Control Interfaces via Real-time Motion Planning and Optimization*. Cornell University
- 2022 *Generating Accurate, Feasible, and Coordinated Bimanual Robot Motions in Real-time*. Workshop on Bimanual Manipulation, ICRA
- 2021 *Methods and Applications for Generating Accurate and Feasible Robot-arm Motions in Real-time*. KavrakiLab, Rice University
- 2021 *Methods and Applications for Generating Accurate and Feasible Robot-arm Motions in Real-time*. Talking-Robotics Series

Selected Teaching

CPSC 487/587 3D Spatial Modeling and Computing, Yale University *course I designed*
 Several areas of computer science and related fields must model and compute how objects are situated in three-dimensional space over time... This course will teach students how to computationally model the spatial configuration of and spatial relationships between objects over time. — Spring 2024. Course Rating: 4.5/5.0

CPSC 485/585 Applied Planning and Optimization, Yale University *course I designed*
 This course introduces students to concepts, algorithms, and programming techniques pertaining to planning and optimization... — Spring 2023. Course Rating: 4.5/5.0, Fall 2023. Course Rating: 4.5/5.0

CPSC 685 Topics on Robot Motion Generation, Yale University *course I designed*
 This course focuses on concepts, approaches, and algorithms related to robot motion generation. Students will read, summarize, present on, and discuss papers and textbook chapters related to search-based path planning, sampling-based path planning, inverse kinematics, and trajectory optimization. — Fall 2022. Course Rating: 4.8/5.0

CPSC 472 Intelligent Robotics, Yale University *Guest Lecture* — Fall 2022

CS/ Psych 770 Human-Computer Interaction, University of Wisconsin-Madison *Guest Lecture* — Spring 2020

Selected Advising

Ph.D. Students

2023 - current	Liam Merz Hoffmeister Xiatao Sun
2024 - current	Chen Liang TJ Vitchutripop Peter Wang
2025 - current	Roshan Klein-Seetharaman

Masters Students

2024 - 2025	Alan Li Yinliang Chen
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Selected Service

2025-Current	Associate Editor , Transactions on Human-Robot Interaction (THRI)
2024	Program Committee Member , HRI
2024	Associate Editor , ICRA
2021	Session Chair , ICRA session Optimization-Based Motion Planning
2021-Current	Review Editor , Frontiers in Robotics and AI

Selected Media

Yale News — *Grasping the future with a robotic arm-hand combo*

Techcrunch — *This robot learns its two-handed moves from human dexterity*

Tech Xplore — *Shared control allows a robot to use two hands working together to complete tasks*

Cosmos, The Science of Everything — *Breaking: robot makes breakfast*

Milwaukee Journal Sentinel — *UW team designs robot hands that work together*

Technical Skills

Programming	Rust, Python, C++, C, C#, Java, OpenGL, ROS, MATLAB, JavaScript, HTML, CSS, WebGL
Software	Blender, 3dsMax, Unity, MotionBuilder, Photoshop, Illustrator, Premier Pro, After Effects, Maya, MudBox, Office